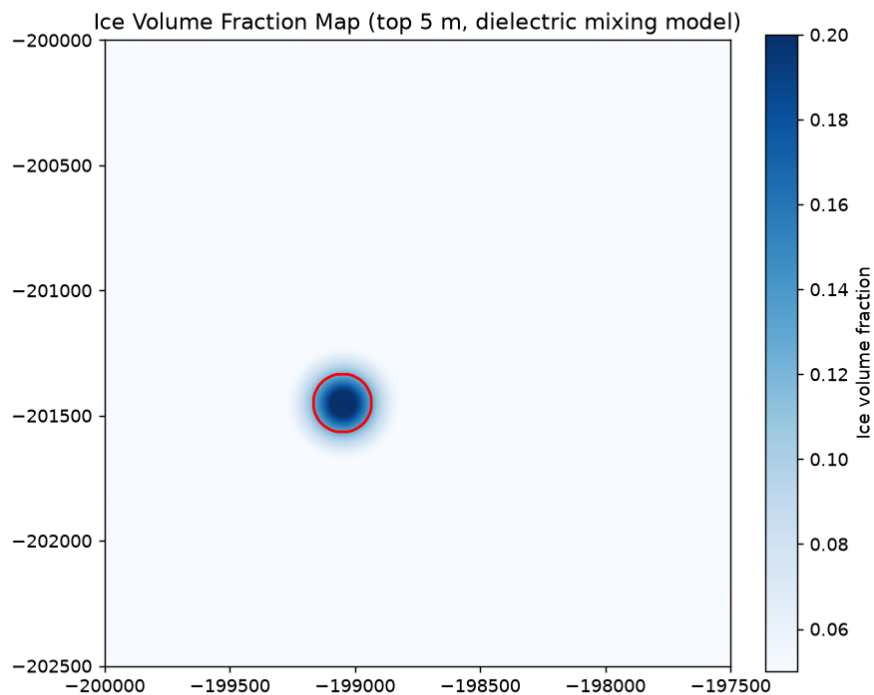


Subsurface Ice Volume Report

Doubly Shadowed Crater - Lunar South Pole

1. Ice Zone Map



2. Volume Estimate

Parameter	Value
Ice-bearing surface area	43,525 m ²
Depth considered	Top 5 m of regolith
Mean ice volume fraction (in zone)	16.5%
Assumed fraction range	5% - 20%
Total estimated ice volume	35,808 m ³
Assumed ice density	917 kg/m ³
Total estimated ice mass	32,836 tonnes

3. Methodology

Ice volume fraction per pixel is estimated using a simplified dielectric mixing-model assumption, scaling radar-derived ice probability (range 0-1) linearly into a plausible volume-fraction range of 5%-20%, consistent with regolith dielectric constant (~3.0) versus water-ice dielectric constant (~3.2). Volume is computed as: $Volume = \sum (pixel_area \times depth \times ice_volume_fraction)$ over all pixels within the ice mask. In an operational pipeline, ice volume fraction would instead be derived

directly from Person 2's CPR/DOP-calibrated radar backscatter inversion rather than this probability-scaled approximation.

4. Key Assumptions & Caveats

(1) Ice is assumed uniformly mixed within the top 5 m, with no stratification. (2) Ice volume fraction is approximated from probability rather than a calibrated dielectric inversion. (3) No ground-truth validation (e.g., from landed instruments) is available at this stage. These caveats should be resolved with refined radar inversion models and, where possible, in-situ validation in later mission phases.